

APPLICATION

BSG-BATS: An open-source data annotation portal and classifier for European bat vocalizations

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Abstract

1. Bats are ecologically important mammals whose monitoring increasingly relies on acoustic data. However, many tools for bat call identification remain subscription-based, closed-source, region-specific or limited in scalability, creating barriers to global data integration and method development.
2. We present BSG-BATS, an open-access annotation portal and convolutional neural network (CNN)-based classifier for bat calls. The portal enables researchers and practitioners to annotate calls at species or phonic group level and contribute directly to the iterative improvement of the classifier. As a proof of concept, we trained the first BSG-BATS model using over 4000 annotated recordings of 21 European species. It gave promising results and outperformed many commonly used commercial and academic tools by providing higher AUC (area under the receiver operating characteristic curve) scores for test data. All annotations are transparently annotated in the portal (<https://bsg.laji.fi/bats/identification/instructions>) and the trained model, code and documentation are available at www.zenodo.org/records/15495676.
3. BSG-BATS offers a foundation for community-driven development of bat sound identification tools. By integrating annotation and model retraining on an open-access platform, it enables collective improvement and adaptation to new regions, species and sound types. We invite all bat researchers and experts to join this collaborative effort, whether by contributing data (bsg-bat@helsinki.fi), annotating sounds (<https://bsg.laji.fi/bats/identification/instructions>) or testing the model in their own work (<https://www.zenodo.org/records/15495676>).

Katarina Meramo and Panu Somervuo shared first authors.

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KEYWORDS

bat sound recognition, bioacoustics, Chiroptera, collaborative research, community science, convolutional neural networks, deep learning, echolocation

1 | INTRODUCTION

Understanding the impacts of global change on biodiversity is urgently needed to guide effective conservation efforts. While some taxonomic groups, such as birds, benefit from existing, long-term monitoring programmes, many others remain underrepresented—often due to their elusive behaviour or lower public visibility. Bats (Mammalia: Chiroptera) are a prime example: They are globally distributed, ecologically diverse and play vital roles in ecosystems by providing services such as pollination, seed dispersal and pest control (Aziz et al., 2021; Castillo-Figueroa, 2020; Kasso & Balakrishnan, 2013). Moreover, bats are considered particularly sensitive to environmental changes due to their life-history strategy, including slow reproduction, long lifespan and hibernation behaviour (Erickson & West, 2002; Frick et al., 2020; Jones et al., 2009; True et al., 2021). This vulnerability underscores the need for robust, long-term monitoring to track population trends and emerging threats. However, traditional monitoring approaches, such as mist-netting and roost surveys, are often labour-intensive, inefficient for most species (especially forest-dwelling bats) and may cause disturbance to animals.

Over the last few decades, passive acoustic monitoring has emerged as a non-invasive alternative to traditional methods, enabling long-term data collection across vast geographic areas by recording bat echolocation calls (e.g. Biffi et al., 2024; Hoggatt et al., 2024; Kotila et al., 2023; Revilla-Martín et al., 2021; Thomas & Davison, 2022). The increasing availability of low-cost ultrasound detectors (e.g. Hill et al., 2018) has further accelerated this shift, supporting the growth of citizen science initiatives and generating enormous acoustic data sets. These recordings can be analysed by bioacoustic classification tools to infer species presence and activity patterns. However, the success of this approach is critically dependent on the accuracy and accessibility of the identification software used. Many of the currently existing bat call acoustic classifiers are closed source, region-specific or limited in performance, particularly in distinguishing between acoustically similar species. In addition, the limited user-friendliness of many classifiers hinders their broader adoption, for example, in citizen science projects. Automated acoustic classifiers offer the potential for high-performance monitoring and are increasingly used across ecological studies, but their reliability in real-world settings is not always well validated. Misidentifications can lead to misleading ecological conclusions or ineffective conservation actions (Russo & Voigt, 2016; Rydell et al., 2017).

Automatic classification of bat calls is technically challenging due to the high variability of bat vocalizations within and between species (Frick et al., 2020; Obrist et al., 2004; Russo & Voigt, 2016). In addition to the challenge of distinguishing closely related taxa

based on their echolocation calls, many bat species also produce a range of social vocalizations, feeding and drinking buzzes, which further increases the acoustic complexity of their soundscapes (Bohn & Smotherman, 2023). Additional variation arises from geographic variation in calls, recording conditions, environmental conditions and differences among individuals (Chen et al., 2009; Jiang et al., 2013; Kingston et al., 2001; López-Baucells et al., 2018; Montauban et al., 2021; Sun et al., 2013). Recordings may also contain overlapping calls or non-bat ultrasonic sounds (e.g. insects, rain, mechanical noise), and furthermore, signal quality may degrade due to environmental factors or the inherent technical capabilities of the recorder in use. To perform reliably, classifiers must distinguish similar species while filtering background and ambiguous signals.

Besides these ecological and technical challenges, a key prerequisite for developing an automatic bat call classifier is the availability of a representative data set of species-specific vocalizations. These reference calls set the limit for the patterns that can be expected to be identified using an automatic computer-based identification pipeline, and such data sets are often limited in size and geographic coverage. Recent advances in machine learning, particularly deep learning, have enabled progress in automatic species identification by leveraging convolutional neural networks and data augmentation. In the context of bat sound identification, an increasing number of studies have explored deep learning-based approaches (Bellafkir et al., 2022; Chen et al., 2020; Dierckx et al., 2022; Khalighifar et al., 2022; Kobayashi et al., 2021; Mahbub et al., 2024; Martínez Balvanera et al., 2025; Paumen et al., 2022; Schwab et al., 2023; Tabak et al., 2022; Vogelbacher et al., 2023; Zualkernan et al., 2020), demonstrating the rapidly growing interest and development in the field.

Here, we present BSG-BATS, a community-driven platform that combines an open-access annotation portal with a convolutional neural network (CNN)-based classifier. The portal enables collaborative annotation of bat recordings, while the classifier provides a proof-of-concept model currently covering 21 European bat species. Unlike many existing bioacoustic classification tools, it is open-source, fully transparent and designed to evolve through community contributions. We describe the methods required to produce the classifier and test its performance against a selection of widely used bat call analysis tools. To encourage collaborative development, we introduce a user-friendly Bat Sounds Global annotation portal where researchers and acoustic experts can contribute and annotate bat recordings. Each contribution helps refine the freely available BSG-BATS model and expand its geographic and taxonomic coverage, bringing us closer to accurate, scalable tools for global bat monitoring.

2 | THE BSG-BATS PORTAL

2.1 | Data sources and annotation procedure in BSG-portal

To generate high-quality training data for our bat call classification model, we used the Bird Sounds Global (BSG) annotation portal (<https://bsg.laji.fi/bats/identification/instructions>), which was originally developed within the LIFEPLAN research project (Hardwick et al., 2024) for the validation and annotation of bird sounds. The portal consists of two main modules: one for validating species-specific acoustic templates and another for annotating soundscapes with species labels. For the purposes of this study, we extended the BSG platform to enable annotation of pre-identified bat echolocation calls. While the bird annotation interface is open to the broader expert community, the bat-specific version was originally restricted to a single expert annotator (Katarina Meramo) to ensure consistent data quality. This bat annotation functionality was developed specifically for this project and is now open for all annotators (see Appendix S1 for instructions and the link <https://bsg.laji.fi/bats/identification/instructions> for more detailed descriptions). The portal provides a transparent and collaborative environment. Annotation progress can be followed in real time, and all contributions are visible to the community. This ensures both accountability and motivation for collaborative effort.

Figure 1 illustrates the community-based workflow for developing and using the BSG-BATS model. The numbered steps represent the key stages of this iterative process: (1) researchers and practitioners contribute their own pre-identified bat recordings, (2) recordings are annotated in the BSG-BATS portal, (3) the accumulated annotations are used to retrain and improve the classifier, (4) the updated model is then applied to identify bat species across different regions using freely available tools (the current version is openly accessible via Zenodo) and (5) user feedback from model use helps to further improve the continuous collaborative cycle. Additional details on the annotation process and data contribution are provided in Appendix S1.

2.2 | Data contribution

The audio material used for annotation consisted of pre-identified bat echolocation recordings collected from several complementary sources. These included open-access databases such as Xeno-canto (xeno-canto.org) and the Animal Sound Archive of the Museum für Naturkunde Berlin, as well as unpublished field recordings contributed by collaborating bat researchers from across Europe. The recordings contributed by individual collaborators will be made openly available on xeno-canto, in line with the platform's terms of use. This ensures full attribution while facilitating reuse and transparency in line with open science principles. Notably, large

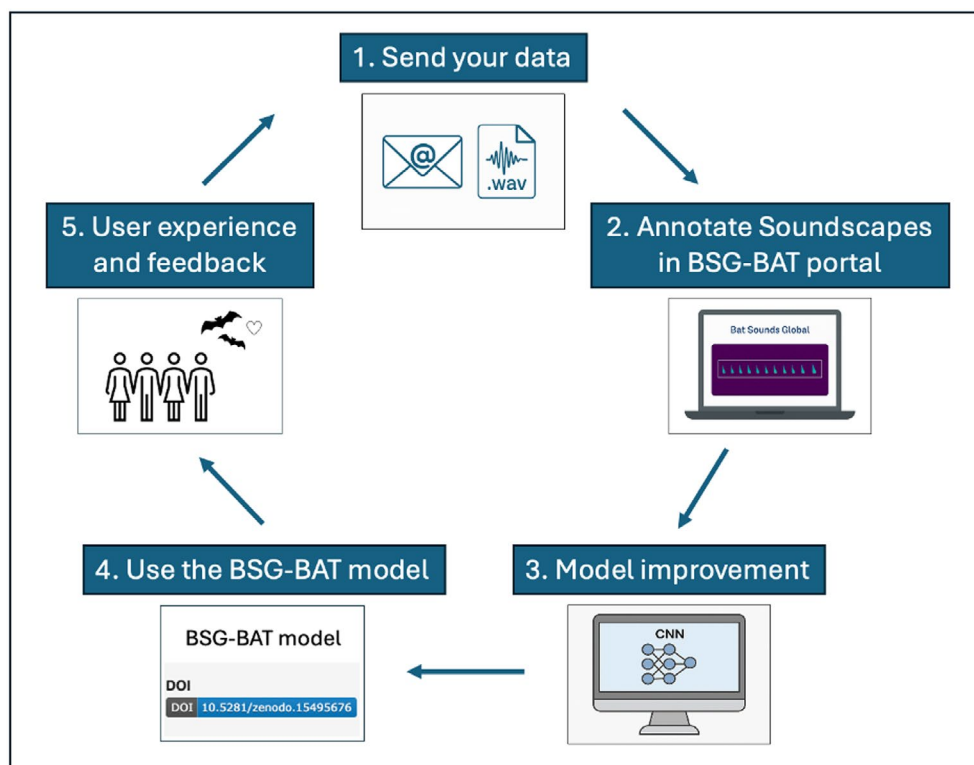


FIGURE 1 Community-based workflow for developing and using the BSG-BATS model. Researchers, students and volunteers can contribute pre-identified bat recordings, annotate soundscapes in the BSG-BATS portal and provide feedback from model use to support continuous improvement and retraining.

portions of the annotated data set were contributed by researchers in Finland, Spain and Italy. The recordings were produced using a range of passive acoustic monitoring devices (e.g. AudioMoth; Open Acoustic Devices and Song Meter SM4 Acoustic Recorder; Wildlife Acoustics), under varying environmental conditions across seasons.

2.3 | Annotation workflow

A total of 4042 annotations were produced using the BSG-BATS platform. Each annotation corresponds to a 10-s audio segment, within which all identifiable bat calls were labelled to the lowest possible taxonomic level (species) or to a phonic group (meaning a set of species that cannot be reliably distinguished based on their echolocation calls alone). While no strict annotation criteria were enforced, only calls with sufficient clarity to enable confident identification (either to species or phonic group level) based on spectrograms, audio and associated metadata were included. In cases where bat calls were audible, but the audio quality was too poor to allow reliable identification to species or phonic group level, the segment was marked as 'poor quality' and later excluded from the training data set to prevent uncertainty. Non-bat sounds such as insects or mechanical noise were also annotated within these segments. These were later labelled as 'background'. They play a crucial role in teaching the classification model to discriminate between relevant bat signals and irrelevant environmental noise.

TABLE 1 Number of annotated audio segments used for training and testing across bat species. The first column lists the species. The second column shows the number of 10-s annotated audio segments used to train the convolutional neural network classifier. The third column shows the number of 1-s windows derived from those annotated 10-s segments. These windows were split into six mutually exclusive groups, and six models were trained so that no 1-s test window was ever included in the training or validation data of the corresponding model. The final column shows the number of independent 1-s audio segments used for testing. Each test segment contains a vocalization of the corresponding species.

Species	Annotated segments (10 s)	Annotated windows (1 s)	Test segments (1 s)
<i>Barbastella barbastellus</i>	181	318	100
<i>Eptesicus nilssonii</i>	495	3501	100
<i>Eptesicus serotinus</i>	165	579	100
<i>Hypsugo savii</i>	233	669	100
<i>Miniopterus schreibersii</i>	167	261	100
<i>Myotis alcaethoe</i>	43	68	68
<i>Myotis crypticus</i>	53	77	77
<i>Myotis daubentonii</i>	191	405	100
<i>Nyctalus leisleri</i>	177	452	100
<i>Nyctalus noctula</i>	151	758	100
<i>Pipistrellus kuhlii</i>	174	539	100
<i>Pipistrellus nathusii</i>	171	964	100
<i>Pipistrellus pipistrellus</i>	199	513	100
<i>Pipistrellus pygmaeus</i>	167	560	100
<i>Plecotus auritus</i>	42	39	39
<i>Plecotus austriacus</i>	83	120	100
<i>Rhinolophus euryale</i>	229	226	100
<i>Rhinolophus ferrumequinum</i>	222	563	100
<i>Rhinolophus hipposideros</i>	256	452	100
<i>Tadarida teniotis</i>	217	792	100
<i>Vespertilio murinus</i>	60	140	100

In total, the annotated data set includes 21 European bat species and five phonic groups, covering a broad geographical and acoustic range. However, because the phonic group annotations were still preliminary and relatively few, only species-level annotations were used in our benchmarking experiments. A summary of annotated species and annotation counts is provided in [Table 1](#).

2.4 | The BSG-BATS classifier

The BSG-BATS classifier was implemented in Python using PyTorch (Paszke et al., 2019) and librosa (McFee et al., 2015). General information regarding deep learning and convolutional neural networks (CNNs) can be found in textbooks such as Chollet (2017). The input to the model is a 1-s spectrogram window, and the classification of the entire audio recording is done by sliding a 1-s analysis window along the duration of the recording with 0.5-s time steps. The BSG-BATS classifier is a CNN whose architecture consists of six convolutional layers followed by two fully connected layers. There is a max-pooling layer after each convolutional layer. The output layer produces confidence scores for species classifications (including a background class). Classification is done in a multi-label fashion so that any number (zero, one or more) of species can be present at the same time in the 1-s analysis window. A full description of the processing steps, CNN architecture and model training is provided in [Appendix S2](#).

We trained the model with an NVIDIA A30 GPU with four CPU workers. GPU memory usage during training was 2832MB and one training epoch took on average 17 s. With the current amount of training data and 50 training epochs, it took, therefore, 15 min to obtain a new model. Adding new data will increase the training time linearly. For classifying new data with a laptop using a single CPU core (12th Gen Intel(R) Core(TM) i5-1235U), it takes less than 1 s for a 10-s WAV file. Memory usage is 500MB and it remains constant with multiple WAV files when WAV files are classified sequentially. With multiple CPU cores, the user can launch a separate classification task for each core (naive parallelization) to speed up the processing of a large number of audio files.

With this workflow, we trained the first proof-of-concept classifier using BSG-BATS portal annotations, currently covering 21 European bat species. This demonstrates that community-contributed data can be transformed into a functional classifier—similar in principle to leading open-source avian classifiers, such as BirdNET (Kahl et al., 2021), which have been primarily trained on community-collected recordings with species-level labels from platforms such as xeno-canto and the Macaulay Library. However, BSG-BATS differs by providing annotations with bounding boxes (providing strong labels), which substantially increase their utility for classifier development and enable more fine-grained training.

3 | BENCHMARKING AGAINST EXISTING CLASSIFIERS

Test data were collected by sampling 100 annotated recordings for each bat species and then randomly selecting a 1-s segment in each, containing the vocalization of the bat. Where we had access to less than 100 recordings of a bat species, we selected multiple non-overlapping 1-s segments from each recording. For three species, the number of annotated recordings was so limited that the number of 1-s segments selected was below 100. Overall, the total number of 1-s test segments covering the 21 target bat species was 1984.

To facilitate fair comparison among the classifiers, we ensured that the test data were non-overlapping with the training and validation data used to parameterize BSG-BATS.

A range of automated tools exists for bat call classification, each with different strengths, limitations, costs and levels of accessibility. Many widely used commercial softwares, such as Kaleidoscope Pro (version

5.6.8; Wildlife Acoustics, [n.d.](https://www.wildlifeacoustics.com/products/kaleidoscope-pro); www.wildlifeacoustics.com/products/kaleidoscope-pro), SonoChiro (version 4; Biotope; www.sonochiro.biotope.fr/en), bcAdmin4 (ecoObs GmbH, [n.d.](https://www.ecoobs.com); www.ecoobs.com) and the BTO Acoustic Pipeline (British Trust for Ornithology; www.bto.org/our-science/products-and-technologies/bto-acoustic-pipeline), offer user-friendly interfaces and fast, pretrained classification models. For example, Kaleidoscope Pro includes graphical tools for both automated and manual review and provides multiple classification modes (conservative, neutral, sensitive) to adjust the balance between precision and recall. However, its underlying algorithms are proprietary and optimized for specific regional species pools. SonoChiro® ([n.d.](https://www.sonochiro.biotope.fr/en)), developed primarily for European species, offers a species-specific model but lacks transparency in its training data and methods, making it difficult to adapt for broader applications. In addition to European species, the software includes a limited number of bat species from the Amazon Basin, North Central America and the Lesser Antilles. BTO's pipeline, used in structured monitoring schemes like the UK's National Bat Monitoring Programme, is not openly documented and relies on a credit-based pricing model. bcAdmin4 is used by consultants and researchers, especially in Central Europe. It includes an efficient on-device classifier built on Apple's CoreML, but as with other commercial software, its algorithms are proprietary and not openly accessible. In our benchmarking, however, it was excluded from the quantitative comparison due to the limited number of consistent species-level identifications, with a substantial portion of outputs remaining at the group level.

In contrast, open-source classifiers provide greater transparency and flexibility. Tadarida (Bas et al., 2017) is an open-source pipeline that incorporates machine learning techniques and allows for flexible customization. However, it is less accessible to users without coding experience, as it requires preprocessing. BatDetect (Aodha et al., 2018) is another free and open classifier that was one of the first to offer bat call detection based on deep learning. Its successor BatDetect2 (Aodha et al., 2022) is able to identify species, but the current model can only identify 17 UK bat species. BatDetect2 is an example of modern classifiers utilizing recent developments in machine learning. Similarly, as our model, BatDetect2 is a CNN-based classifier, but its details are different. BatDetect2 is based on U-net architecture and attention mechanism. It segments calls using object detection before classifying whereas our model does not do any object detection within a 1-s analysis window. An overview of these tools and their main attributes is provided in Table 2 for clarity.

Tool	Availability	Algorithm	Species coverage
BSG-BAT	Open-source	CNN	21 European species
Kaleidoscope Pro	Commercial	Unknown (proprietary)	NA, EU, Southern Africa, Neotropics
SonoChiro	Commercial	Unknown (proprietary)	Mainly EU + limited tropical
BTO Pipeline	Commercial	Unknown (proprietary)	Mainly UK
Tadarida	Open-source	Random forest	Variable
Batdetect2	Open-source	U-Net + attention	17 UK species

TABLE 2 Key characteristics of the bat call classification tools included in our benchmark comparison. Tools are compared by their availability, algorithm type and the approximate taxonomic coverage. Species coverage is indicative and reflects the version available at the time of testing.

For all software, only identifications of 21 target species were taken into account. If the outputs contained additional species, they were ignored and thus did not deflate precision rates. SonoChiro (Biotope) produced only one species label with a confidence score per test recording. Kaleidoscope Pro (Wildlife Acoustics) was run in three different modes: conservative, neutral and liberal. For our test data, the liberal model produced the best results. We used the AUTO ID label and MATCH_RATIO value for its confidence score. Tadarida (Bas et al., 2017) outputs several species labels with confidence scores. Notably, we used the only Tadarida version currently available online, which corresponds to the original implementation published in 2017. While the authors have since substantially improved the software, these updated versions are not openly accessible. BTO Acoustic Pipeline (British Trust for Ornithology) and BatDetect2 both give species labels and their confidence scores as output. BSG-BATS also gives a confidence value for each identification and may provide multiple species labels in case of overlapping vocalizations.

Based on species detections and their confidence scores, we calculated species-specific AUC (area under the receiver operating characteristic curve) score for each method. AUC can be interpreted as a probability for a classifier producing a higher confidence score for a randomly selected example from the target species compared to the confidence score for a randomly selected example from another species, that is, a higher value is better.

To evaluate practical usability, we also compared the processing times used by the respective classifiers. These tests were performed on same-sized data sets (1984 files). In conducting these comparisons, we note that direct comparisons are limited by differences in hardware, classifier design and preprocessing steps, and that individual pieces of software have not been optimized for speed. Nonetheless, we feel that the results still provide useful context for performance expectations.

The comparison among the classifiers shows that BSG-BATS yielded the highest AUC scores averaged over all species (Appendix S3). Concerning species-specific results, BSG-BATS yielded the best performance for most species (Figure 2; Appendix S3), BatDetect2 performed well for the 14 species it was able to identify and BTO was the third best method.

Concerning computational performance, BSG-BATS processed the full data set in approximately 2 min on a Linux laptop using four CPU cores (3 min with a single core). This demonstrates that the lightweight CNN architecture can run efficiently even on a standard desktop computer. Because this is the most likely way most biologists will use the software, we focused our benchmarking on CPU performance. BatDetect2 used 10 min with four CPU cores (16 min with a single core). Kaleidoscope Pro and the BTO Acoustic Pipeline were both run on a 2024 MacBook Air. Kaleidoscope completed classification in 11.7–13.2 s, depending on the selected mode (conservative, neutral, sensitive). The BTO Pipeline took about 7 min 45 s to upload the batch, and the total time from upload to receiving the results was 1 h 8 min. SonoChiro (using the Mediterranean classifier, as including all European species), run on an HP EliteBook, required approximately 8 min. Tadarida's runtime after the preprocessing steps was 8 min 29 s.

4 | COMMUNITY-DRIVEN DEVELOPMENT AND NEXT STEPS

With BSG-BATS, our goal is not only to provide a classifier but also to establish an open framework for continuous improvement. The current model (21 European species) demonstrates what can be achieved with portal-based annotations, but its true potential lies in expanding coverage through community participation.

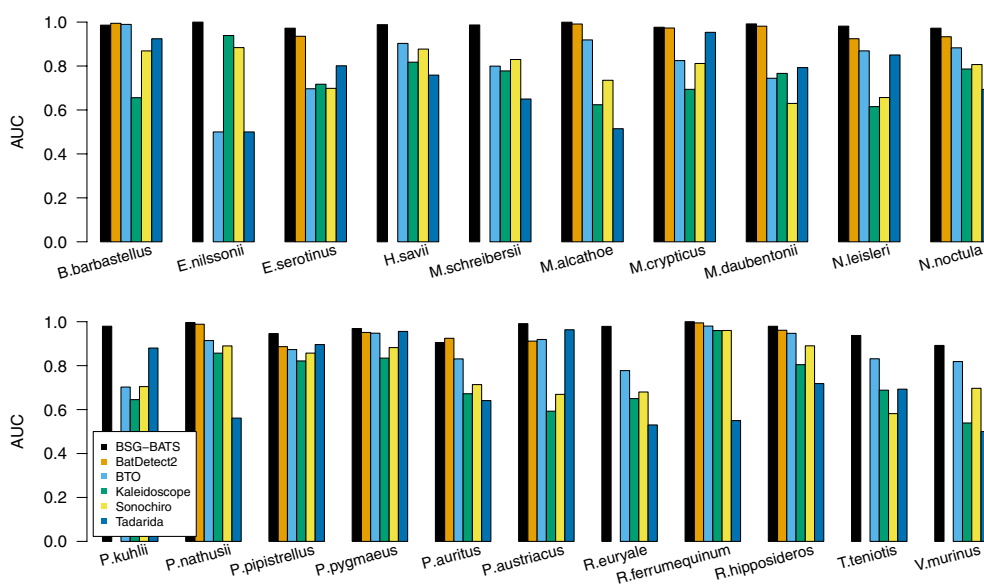


FIGURE 2 Benchmark performance of BSG-BATS and existing classifiers. Each bar shows the species-specific AUC for a given classifier, with different colours indicating different classifiers (as shown in the figure legend). For some species, performance may appear optimistic due to partial similarity between the training and testing data sets used in this study. AUC, area under the receiver operating characteristic curve.

A key novelty is the integration with the BSG-BATS portal, where data can be contributed, annotated and transparently tracked in real time. This offers clear advantages compared with closed or commercial pipelines: All annotations remain visible, anyone can participate and the model can be retrained openly as new data accumulate. Annotation can be done at the species or phonic-group level, and annotators can indicate certainty levels, helping to manage uncertainty. This model of community-driven data collection and curation has already proven effective in the context of bird sound annotation through the Bird Sounds Global portal (Lauha et al., 2022; Lehtikoinen et al., 2023). Guidelines for annotation, data sharing and participation are provided in [Appendix S1](#).

Future development requires several parallel efforts. These include expanding taxonomic and regional coverage, since annotated data from new species and areas are crucial, and improving background classes by adding more diverse non-bat sounds (e.g. insects, rain, mechanical noise) to reduce false positives in real-world conditions. Strengthening phonic groups is also important, as grouping acoustically similar species offers a practical and ecologically meaningful solution widely used in large-scale monitoring, such as the Bat Monitoring Programme in south-western Europe (Torre et al., 2021).

It will also be important to ensure greater geographic diversity in the data, as exceptionally high classification scores for some species may partly reflect limited variation in the training and test data sets—for instance, many *Eptesicus nilssonii* recordings originated from the same contributor (BatLab Finland), which may introduce shared characteristics and exaggerate performance. Broader geographic and acoustic diversity will be critical to strengthen generalization in future iterations. Finally, we aim to address potential non-independence in evaluation: For species with limited data, multiple non-overlapping 1-s segments from the same recording were used to reach sufficient test sample sizes, which may slightly inflate performance estimates. However, this affects only a few species with scarce data, while the majority of test segments originated from different recordings and contributors. We will aim to reduce this limitation in future iterations by prioritizing the collection of more independent recordings per species.

By design, the portal allows these contributions to feed back into retraining, creating a cycle of collaborative improvement (Figure 1). The current model is openly available via Zenodo (Somervuo, 2025; <https://doi.org/10.5281/zenodo.15495676>), and we welcome researchers, students and practitioners to test it, provide feedback and share their own data and annotations.

5 | CONCLUSIONS

Monitoring biodiversity under rapid global change requires scalable, accurate and transparent acoustic classifiers. With BSG-BATS, we provide not only one of the first open-source classifiers for European bats but, more importantly, an open framework that grows stronger through collaboration. The BSG-BATS portal enables the community to contribute data, annotate recordings and directly shape future models. This collaborative cycle allows the classifier to

expand beyond Europe, adapt to new taxa and connect with global efforts. In doing so, it builds on the growing movement towards open and community-driven bioacoustic classifications (e.g. Aodha et al., 2018; Martínez Balvanera et al., 2025; Mahbub et al., 2024).

We warmly invite colleagues across regions and disciplines to join us—by sharing data, annotating sounds, testing models or providing feedback. Together, we can build a global, sustainable and open tool for bat bioacoustics that advances science, conservation and biodiversity monitoring.

AUTHOR CONTRIBUTIONS

Katarina Meramo, Panu Somervuo, Otso Ovaskainen and Thomas M. Lilley designed the study. Meeri Rannisto managed the development of the BSG annotation portal. Katarina Meramo annotated the training data using the BSG portal. Katarina Meramo, Thomas M. Lilley, Adrià López-Baucells and Fabrizio Gili contributed acoustic data. Panu Somervuo and Meeri Rannisto handled data processing and management. Panu Somervuo developed the classification model. Katarina Meramo and Panu Somervuo led the writing with input from all authors. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest to declare.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210x.70220>.

DATA AVAILABILITY STATEMENT

The trained BSG-BATS classification model and all associated code are archived at Zenodo: Data available via <https://doi.org/10.5281/zenodo.15495676> (Somervuo, 2025). All original and annotated audio recordings contributed by collaborators will be openly available on Xeno-canto (<https://xeno-canto.org>) in accordance with the platform's terms of use.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. Instructions for annotation and data contribution.

Appendix S2. BSG-BATS classifier.

Appendix S3. Method comparison results.

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